MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY CHAPTER TEST – 02 (PROBABILITY)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 14: Probability** | **Max. Marks: 20**

BOARD-STYLE WORKED SOLUTIONS | Step-by-Step Solutions | CBSE Class X Mathematics | Academic Session 2026–27

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

Question 1 [1 Mark]

If $P(E) = 0.05$, what is the probability of 'not E' (i.e. $P(\bar{E})$)?

(a) **0.95** (b) **0.05**

(c) **1.05** (d) **0.095**

Working & Explanation:

- By the property of complementary events, for any event E and its complement $\bar{E}$ (not E):

$$P(E) + P(\bar{E}) = 1 \implies P(\bar{E}) = 1 - P(E)$$

- Substituting the given value $P(E) = 0.05$:

$$P(\bar{E}) = 1 - 0.05 = 0.95$$

CORRECT ANSWER: OPTION (a) ⟶ 0.95

Question 2 [1 Mark]

A single die is thrown once. The probability of getting a prime number is:

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$

(c) $\frac{2}{3}$ (d) $\frac{1}{6}$

Working & Explanation:

- When a standard die is thrown once, the sample space is:

$$S = \{1, 2, 3, 4, 5, 6\} \implies n(S) = 6$$

- The prime numbers in the sample space are 2, 3, and 5 (Note: 1 is neither prime nor composite):

$$E = \{2, 3, 5\} \implies n(E) = 3$$

- Therefore, by the classical probability formula:

$$P(\text{prime number}) = \frac{n(E)}{n(S)} = \frac{3}{6} = \frac{1}{2}$$

CORRECT ANSWER: OPTION (b) ⟶ $\frac{1}{2}$

Question 3 [1 Mark]

From a standard deck of 52 cards, the probability of drawing a queen of clubs or a king of hearts is:

(a) $\frac{1}{13}$ (b) $\frac{1}{26}$

(c) $\frac{1}{52}$ (d) $\frac{2}{13}$

Working & Calculation:

- Total cards in a standard deck: $n(S) = 52$.
- In a standard deck, there is exactly 1 Queen of Clubs and exactly 1 King of Hearts.
- Favourable outcomes $n(E) = 1 + 1 = 2$:

$$P(\text{Queen of Clubs or King of Hearts}) = \frac{n(E)}{n(S)} = \frac{2}{52} = \frac{1}{26}$$

CORRECT ANSWER: OPTION (b) $\longrightarrow \frac{1}{26}$

Question 4 [1 Mark]

Cards numbered from 1 to 20 are mixed thoroughly and a card is drawn at random. The probability that the drawn card bears a number which is a multiple of 3 is:

(a) $\frac{3}{10}$ (b) $\frac{1}{3}$

(c) $\frac{3}{20}$ (d) $\frac{7}{20}$

Working & Calculation:

- Total cards numbered from 1 to 20: $n(S) = 20$.
- Multiples of 3 in $\{1, 2, \ldots, 20\}$ are $E = \{3, 6, 9, 12, 15, 18\}$:

$$n(E) = 6 \implies P(\text{multiple of 3}) = \frac{n(E)}{n(S)} = \frac{6}{20} = \frac{3}{10}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{3}{10}$

Question 5 (Assertion–Reason) [1 Mark]

Assertion (A): If the probability of winning a game is 0.4, then the probability of losing it is 0.6.

Reason (R): For any event E, $P(E)+P(\bar{E}) = 1$, where $\bar{E}$ represents the negation/complement of event E.

Options:

(a) **Both (A) and (R) are true, and (R) is the correct explanation of (A).**
(b) **Both (A) and (R) are true, but (R) is not the correct explanation of (A).**
(c) **(A) is true, but (R) is false.**
(d) **(A) is false, but (R) is true.**

Step-by-Step Logical Analysis:

1. **Assertion (A):** Let E be winning the game $\implies P(E) = 0.4$. Losing is the complement $\bar{E} \implies P(\bar{E}) = 1 - 0.4 = 0.6$. $\implies$ **(A) is TRUE.**
2. **Reason (R):** The fundamental relation for complementary events is $P(E) + P(\bar{E}) = 1$. $\implies$ **(R) is TRUE.**
3. **Explanation Check:** Assertion (A) directly uses the complement formula given in (R). Thus, (R) correctly explains (A).

CORRECT ANSWER: OPTION (a) $\longrightarrow$ Both (A) and (R) are true, and (R) is the correct explanation of (A).

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [2 Marks]

Question 6 [2 Marks]

Two coins are tossed simultaneously. Write all possible outcomes and find the probability of getting exactly one tail.

Step-by-Step Solution:

- **Step 1: All possible outcomes (Sample Space)**
  When two coins are tossed simultaneously, the complete sample space S is:

$$S = \{\text{HH}, \text{HT}, \text{TH}, \text{TT}\} \implies n(S) = 4$$

- **Step 2: Favourable outcomes for "exactly one tail"**
  Let E be the event of getting exactly one tail:

$$E = \{\text{HT}, \text{TH}\} \implies n(E) = 2$$

- **Step 3: Calculate the probability**

$$P(\text{exactly one tail}) = \frac{n(E)}{n(S)} = \frac{2}{4} = \frac{1}{2}$$

FINAL ANSWER: $\mathbf{P(\text{exactly one tail}) = \frac{1}{2}}$ [Sample Space: {HH, HT, TH, TT}]

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [9 Marks]

Question 7 [3 Marks]

A bag contains 5 red balls, 8 white balls, 4 green balls, and 7 black balls. If one ball is drawn at random, find the probability that it is:

Composition of the Bag: Total number of balls in the bag:

$$n(S) = 5 \text{ (red)} + 8 \text{ (white)} + 4 \text{ (green)} + 7 \text{ (black)} = 24 \text{ balls}$$

(i) A black ball [1 Mark]

Number of black balls in the bag = 7:

$$P(\text{black}) = \frac{\text{Number of black balls}}{n(S)} = \frac{7}{24}$$

Answer (i): $\mathbf{P(\text{black ball}) = \frac{7}{24}}$

(ii) A red or white ball [1 Mark]

Number of red or white balls = $5 + 8 = 13$:

$$P(\text{red or white}) = \frac{\text{Number of red or white balls}}{n(S)} = \frac{13}{24}$$

Answer (ii): $\mathbf{P(\text{red or white}) = \frac{13}{24}}$

(iii) Not green [1 Mark]

Number of balls that are not green = $n(S) - \text{Green} = 24 - 4 = 20$:

$$P(\text{not green}) = \frac{20}{24} = \frac{5}{6} \quad \text{or via complement:} \quad 1 - P(\text{green}) = 1 - \frac{4}{24} = \frac{5}{6}$$

Answer (iii): $\mathbf{P(\text{not green}) = \frac{5}{6}}$

Question 8 [3 Marks]

Cards numbered 1 to 30 are put in a bag. A card is drawn at random from this bag. Find the probability that the drawn card has:

Sample Space: Total cards numbered 1 to 30: $n(S) = 30$.

(i) A prime number greater than 10 [1 Mark]

Prime numbers strictly between 10 and 30 are $E_1 = \{11, 13, 17, 19, 23, 29\}$:

$$n(E_1) = 6 \implies P(\text{prime } > 10) = \frac{n(E_1)}{n(S)} = \frac{6}{30} = \frac{1}{5}$$

Answer (i): $\mathbf{P(\text{prime } > 10) = \frac{1}{5}}$

(ii) A number divisible by both 2 and 3 [1 Mark]

Divisible by both 2 and 3 $\iff$ divisible by LCM$(2, 3) = 6$. Multiples are $E_2 = \{6, 12, 18, 24, 30\}$:

$$n(E_2) = 5 \implies P(\text{divisible by both 2 and 3}) = \frac{n(E_2)}{n(S)} = \frac{5}{30} = \frac{1}{6}$$

Answer (ii): $\mathbf{P(\text{divisible by 2 and 3}) = \frac{1}{6}}$

(iii) A number not divisible by 5 [1 Mark]

Multiples of 5 in $\{1, \ldots, 30\}$ are $\{5, 10, 15, 20, 25, 30\}$ (6 outcomes). Non-multiples $= 30 - 6 = 24$:

$$P(\text{not divisible by 5}) = \frac{24}{30} = \frac{4}{5} \quad \text{or via complement:} \quad 1 - \frac{6}{30} = \frac{4}{5}$$

Answer (iii): $\mathbf{P(\text{not divisible by 5}) = \frac{4}{5}}$

Question 9 [3 Marks]

Five cards—the ten, jack, queen, king, and ace of diamonds—are well-shuffled with their face downwards. One card is then picked up at random.

Initial Card Set: $\{10, \text{J}, \text{Q}, \text{K}, \text{A}\}$ of diamonds $\implies n(S) = 5$ cards.

(i) What is the probability that the card is the queen? [1 Mark]

There is exactly 1 Queen in the 5 cards:

$$P(\text{queen}) = \frac{n(\text{queen})}{n(S)} = \frac{1}{5}$$

Answer (i): $\mathbf{P(\text{queen}) = \frac{1}{5}}$

(ii) If the queen is drawn and put aside, what is the probability that the second card picked is: [2 Marks]

When the Queen is removed, remaining cards are $\{10, \text{J}, \text{K}, \text{A}\} \implies n(S') = 4$ cards.

(a) an ace? [1 Mark]

There is 1 Ace remaining in the 4 cards: $P(\text{ace}) = \frac{1}{n(S')} = \frac{1}{4}$.

Answer (ii)(a): $\mathbf{P(\text{an ace}) = \frac{1}{4}}$

(b) a queen? [1 Mark]

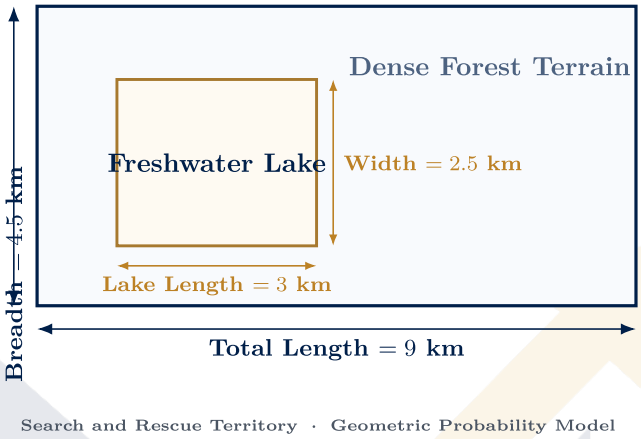
Since the Queen was set aside, 0 Queens remain in the deck (Impossible event): $P(\text{queen}) = \frac{0}{n(S')} = 0$.

Answer (ii)(b): $\mathbf{P(\text{a queen}) = 0}$

SECTION D: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 10: Case Study Analysis [4 Marks]

Case Study Scenario:
A disaster management cell is coordinating search and rescue operations for a missing helicopter that is reported to have crashed somewhere in a rectangular region of length 9 km and breadth 4.5 km. Inside this rectangular territory, there is a freshwater lake of length 3 km and width 2.5 km. The rest of the area is dense forest terrain.



Search and Rescue Territory · Geometric Probability Model

(i) Calculate the total geographical area of the search region. [1 Mark]

$$\text{Total Area} = \text{Length} \times \text{Breadth} = 9 \text{ km} \times 4.5 \text{ km} = 40.5 \text{ km}^2 \quad \left(= \frac{81}{2} \text{ km}^2\right)$$

Answer (i): Total Geographical Area = 40.5 km^2

(ii) Calculate the surface area of the lake. [1 Mark]

$$\text{Area of Lake} = \text{Length} \times \text{Width} = 3 \text{ km} \times 2.5 \text{ km} = 7.5 \text{ km}^2 \quad \left(= \frac{15}{2} \text{ km}^2\right)$$

Answer (ii): Surface Area of the Lake = 7.5 km^2

(iii) What is the probability that the helicopter crashed inside the lake? [1 Mark]

By geometric probability:

$$P(\text{crashed in lake}) = \frac{\text{Area of Lake}}{\text{Total Area}} = \frac{7.5}{40.5} = \frac{75}{405} = \frac{5}{27} \quad \left(\text{or } \frac{15/2}{81/2} = \frac{15}{81} = \frac{5}{27}\right)$$

Answer (iii): P(crashed in lake) = $\frac{5}{27}$

(iv) What is the probability that the helicopter crashed in the forest region outside the lake? [1 Mark]

$$\text{Forest Area} = 40.5 - 7.5 = 33 \text{ km}^2 \implies P(\text{crashed in forest}) = \frac{33}{40.5} = \frac{330}{405} = \frac{22}{27} \quad (\text{or } 1 - \frac{5}{27} = \frac{22}{27}).$$

Answer (iv): P(crashed in forest region) = $\frac{22}{27}$

★★★ END OF WORKED SOLUTIONS · MATHS MANTHAN ★★★